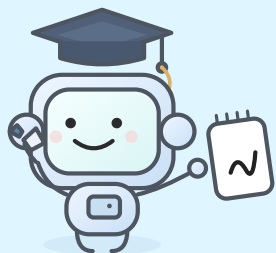


EvalMem in Action: a Worked Example



Q: “When did Alex move into his new apartment?” Gold answer: “14 August 2023”

Encoding Examiner



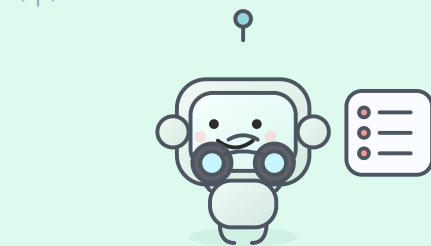
TRACE: Probes the memory store **M**

“ I finally moved into my new apartment yesterday — exhausted but happy! ”

(timestamped 2023-08-15)

Verdict: Exist

Retrieval Examiner



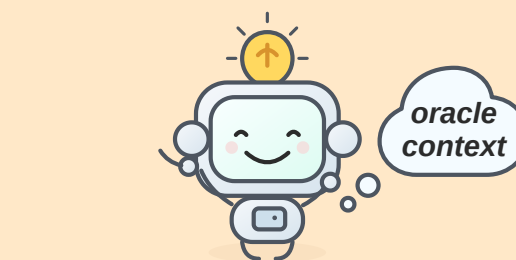
Inspects the system's top-**K**

- “Alex enjoys morning coffee”
- “weekend hiking trip”
- “Alex’s gym routine”
- “...”

evidence is NOT in top-**K**

Verdict: Miss → flags **RF**

Generation Examiner



BYPASSES the memory system;
feeds $Q + C_{\text{oracle}} + A_{\text{gold}}$ to the LLM

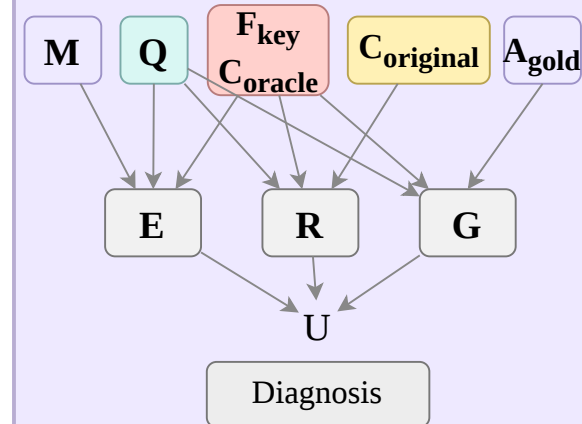
$Q + C_{\text{oracle}}$
 A_{oracle} : 14 August

A_{gold} : “14 August 2023”

Compare A_{gold} and A_{oracle}

Verdict: Correct

“Framework Structure” reference key



Each robot probes a **DIFFERENT** internal target of the memory system, in parallel.

FINAL DIAGNOSIS BOX

Defect set: { **RF** }

The system **DID** write the fact, but its retriever failed to surface it.
End-to-end accuracy alone could not pinpoint this.